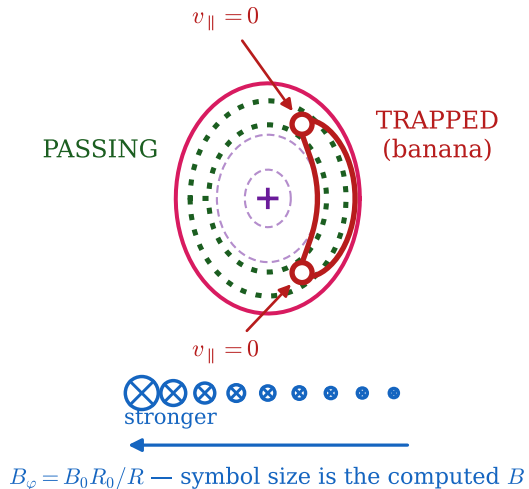


the banana never reaches the inboard side —
that is the whole content of the picture



the mirror condition, in four lines

$$\mu = \frac{mv_{\perp}^2}{2B}, \quad E = \frac{1}{2}mv_{\parallel}^2 + \mu B \quad \text{conserved}$$

$$v_{\parallel} = 0 \Rightarrow B_t = B_{\min} v^2 / v_{\perp 0}^2$$

$$\text{trapped} \Leftrightarrow B_t \leq B_{\max} = B_{\min} \frac{1+\varepsilon}{1-\varepsilon}$$

$$\Leftrightarrow (v_{\parallel 0} / v)^2 \leq \frac{2\varepsilon}{1+\varepsilon}$$

Here $\varepsilon = 0.31$, so the window is $(v_{\parallel 0} / v)^2 \leq 0.47$.

The banana drawn has 0.12; the passing orbit 0.90.

The trapped FRACTION is roughly $\sqrt{2\varepsilon / (1+\varepsilon)} \approx 0.69$ here.
That is not a small correction: a large minority of particles
never sees the inboard side. §37 turns that asymmetry into a
current.

Two exaggerations, stated: ε is drawn larger than a
reactor's, and the banana width is drawn far larger than
 $q\rho / \sqrt{\varepsilon}$, so that the orbit is visible at print size.

Explanatory sketch. The orbit shape is computed from $v_{\parallel}(\theta)$; the inverse aspect ratio and the banana width are exaggerated for legibility.